# Alex Zongo

Ph.D. Candidate in Mechanical and Aerospace Engineering, The George Washington University

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## RESEARCH INTERESTS

**Methods**: robust multi-agent reinforcement learning under degraded or adversarial sensing; runtime safety filtering for learned control policies; conditional flow matching for multi-agent trajectory prediction and generation; language models in the multi-agent decision loop; multi-agent world models for planning.
**Applications**: aircraft autonomy, drones and aerial robots, air traffic control; transferable to driving and mobile robots.

## EDUCATION

**Ph.D., Mechanical and Aerospace Engineering** — Aug 2024 – Present
*The George Washington University* — Washington, DC, USA
Advisor: Prof. Peng Wei. Safe AI-based control theory and algorithms for aircraft autonomy and multi-agent systems.

**M.S., Control Science and Engineering** — Sep 2021 – May 2024
*Tsinghua University, Department of Automation* — Beijing, China
Advisor: Prof. Qing Li. Reinforcement learning for fault-tolerant flight control.

**B.S., Aircraft Design and Engineering** — Sep 2018 – Jun 2021
*Beihang University (Beijing University of Aeronautics and Astronautics), School of Aeronautics* — Beijing, China
Advisor: Dr. Jidong Wang. Aircraft design, flight dynamics and control, aerodynamics.

**Machine Learning Summer School** — Jun 2022 – Aug 2022
*University of Oxford* — Oxford, United Kingdom

## HONORS AND AWARDS

**3rd Best Paper of Conference, Best in Track, and Best in Session, DASC 2026 [C3]** — 2026
**Best in Session Paper Award, ICNS 2026 [C2]** — 2026
**Chinese Government Scholarship (M.S.), Tsinghua University** — 2021 – 2024
**Chinese Government Scholarship (B.S.), Beihang University** — 2018 – 2021
**Outstanding Academic Achievement Award (1st Prize), Beihang University** — 2019, 2020
**Taiwan Ministry of Foreign Affairs Scholarship, NCKU and Fu Jen Catholic University** — 2016 – 2018

## PUBLICATIONS

Conference and workshop papers are peer reviewed. Project pages, slides and posters at alexzongo.com/publications.

### Conference papers

[C7] M. Ghazanfari, M. Casanova, J. Kam, **A. Zongo**, P. Wei, T. Darrell, A. Bayen. *Air Traffic Control Using Large Language Models: Prompt Engineering, Architecture, and Evaluation*. AIAA SciTech Forum, 2027 (to appear). [arXiv] [Project]

[C6] **A. Zongo**, S. Ochoa, L. Sun, A. Nguyen, S. Tran, P. Wei. *Regional Air Mobility Operational Energy Assessment Using a Full-Envelope eVTOL Power Model*. AIAA SciTech Forum, 2027 (to appear).

[C5] **A. Zongo**, F. Fotiadis, U. Topcu, P. Wei. *Robust Multi-Agent Reinforcement Learning for Small UAS Separation Assurance under GPS Degradation and Spoofing*. IEEE Conference on Decision and Control (CDC), 2026 (to appear). Also presented at the ACC 2026 Workshop on Autonomy in Transportation. [arXiv] [Project] [Poster]

[C4] Y. Pang, A. Aziz, **A. Zongo**, P. Wei, K. Crandall. *Large Language Model Guided Reinforcement Learning for Maritime Ship Collision Avoidance*. IEEE OCEANS, Monterey, CA, 2026 (to appear). [Project]

[C3] **A. Zongo**, P. Wei. *Runtime Safety Filtering for Learned Small UAS Separation Policies under GNSS Degradation*. AIAA/IEEE Digital Avionics Systems Conference (DASC), Orlando, FL, 2026 (proceedings forthcoming). 3rd Best Paper of Conference, Best in Track, Best in Session. [arXiv] [Project] [Slides]

[C2] **A. Zongo**, P. Wei. *eVTOL Aircraft Energy Consumption Estimation with Conflict Resolution in High-Density Airspaces*. Integrated Communications, Navigation and Surveillance Conference (ICNS), Herndon, VA, 2026. Best in Session. [Paper] [arXiv] [Project] [Slides]

[C1] **A. Zongo**, Q. Li. *Towards Intelligent Fault-Tolerant Attitude Flight Control of Fixed-Wing Aircraft*. International Conference on Guidance, Navigation and Control (ICGNC), Changsha, 2024. Lecture Notes in Electrical Engineering, vol. 1353, Springer, 2025. [Paper] [Code] [Slides]

### Workshop papers

[W1] I. Sharifi, **A. Zongo**, P. Wei. *Fine-Tuning Large Language Models for Cooperative Tactical Deconfliction of Small Unmanned Aerial Systems*. CVPR Workshop on Multi-Agent Embodied Intelligent Systems (MEIS), Denver, CO, 2026. [Paper] [arXiv] [Project]

### Manuscripts in preparation

[P1] **A. Zongo**, M. Petit, L. Brusset, J. Kam, E. Torun, M. Ghazanfari, P. Wei, A. Bayen. *Aircraft Trajectory Prediction under ATC Instruction Uncertainty*. Manuscript in preparation, 2026. [Code] [Project]

### Other writing

[E1] **A. Zongo**. *Aircraft Traffic Control: Managing Order in a Crowded Sky*. Research essay, 2025. [Zenodo] [Online]

## RESEARCH EXPERIENCE

**Graduate Research Assistant** Sep 2024 – Present
*Intelligent Aerospace Systems Lab, The George Washington University (advisor: Prof. Peng Wei)* Washington, DC

- **Robust MARL.** Modeled GNSS degradation and spoofing as a bounded observation adversary and derived a closed-form, first-order worst-case perturbation from the multi-agent value gradient, so robust separation policies train without adversarial co-training [C5].
- **Runtime safety.** Designed and compared runtime filters that wrap a learned separation policy and act on either its action or its observation, using a shared worst-case state estimate [C3].
- **Energy-aware air mobility.** Coupled an eVTOL power model with MVP conflict resolution in BlueSky, measured the energy cost of deconfliction over ∼72,000 simulated transits (median overhead below 1.5%), and learned per-flight reserve bounds with conservative uncertainty estimates [C2, C6]. NASA University Leadership Initiative.
- **Language models.** Co-developed a simulation-to-language pipeline (∼38k rule-consistent samples) and LoRA fine-tuning (SFT, GRPO) of LLMs for sUAS tactical deconfliction; decision accuracy rose from 27% to 88% [W1].

**Visiting Researcher** May 2026 – Aug 2026
*CITRIS and the Banatao Institute, University of California, Berkeley (host: Prof. Alexandre Bayen)* Berkeley, CA

- Flight trajectory prediction and generation via flow matching from ADS-B and ATC voice data under uncertain communication: accounting for information lost in transcription and for variability in how an ATC instruction is executed [P1].

**Graduate Research Assistant** Sep 2021 – May 2024
*Navigation and Control Lab, Department of Automation, Tsinghua University (advisor: Prof. Qing Li)* Beijing, China

- Developed reinforcement-learning controllers (CEM-TD3, PPO) for fault-tolerant attitude control of a 6-DoF fixed-wing aircraft model under actuator saturation, jamming, and effectiveness loss, wing icing, and a shifted center of gravity; open-sourced the code [C1].

**Robotics Software Engineering Intern** Jun 2022 – Nov 2022
*Popular Robotics* Beijing, China

- Built a biped robot simulation in Gazebo with ROS/ROS 2 and designed a curriculum module on gait fundamentals, simulation, and control.

## TALKS AND PRESENTATIONS

**Embodied AI: Vision-Language-Action Models** Oct 2026
*Invited panel talk, SEINAR 2026, World AI Week* Online

**Runtime Safety Filtering for Learned sUAS Separation Policies under GNSS Degradation** Sep 2026
*Conference talk, DASC 2026* Orlando, FL

**Fine-Tuning LLMs for Cooperative Tactical Deconfliction of Small UAS** Jun 2026
*Poster, MEIS Workshop at CVPR 2026* Denver, CO

**Robust MARL for sUAS Separation Assurance under GPS Degradation and Spoofing** May 2026
*Poster, ACC 2026 Workshop on Autonomy in Transportation* New Orleans, LA

**eVTOL Energy Overhead Estimation under Conflict Resolution in High-Density Airspaces** Apr 2026
*Conference talk, ICNS 2026* Herndon, VA
**Le parcours d'un Burkinabé dans le domaine de l'aéronautique et de l'espace** Apr 2025
*Invited webinar, Initiative Burkinabè pour l'Aéronautique et l'Espace (IBAE)* Online
**Towards Intelligent Fault-Tolerant Attitude Flight Control of Fixed-Wing Aircraft** Aug 2024
*Conference talk, ICGNC 2024* Changsha, China
**Intelligent Control and Reinforcement Learning** Apr 2024
*Academic forum, Beijing Universities AI Forum and Tsinghua Ph.D. Academic Discussion* Beijing, China

## TEACHING AND MENTORING

**Guest Lecturer, Large Language and Vision Models (MAE 6291 / CSCI 6907)** Fall 2026
*The George Washington University*
Lecture: *From Noise to Data: Diffusion Models through Flow Matching*. [Interactive notes]

**Graduate Teaching Assistant, Linear Systems Dynamics (MAE 3134)** Spring 2025
*The George Washington University*
Recitations, office hours, and grading.

**Research Mentor** Summer 2025 – Present
*Intelligent Aerospace Systems Lab, The George Washington University*

- M.S. student: maritime traffic simulation and multi-agent RL for collision avoidance (Fall 2025 – Oct. 2026) [C4].
- Undergraduate student: air traffic simulation and research with BlueSky (Summer 2025).

## PROFESSIONAL SERVICE

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|---|---|
| **Journal reviewer** | IEEE TAES; IEEE T-ITS; Transportation Research Part C; AIAA JGCD; AIAA JAIS; Eng. Appl. of Artificial Intelligence |
| **Conference reviewer** | IEEE CDC; IEEE ICRA; IEEE ITSC; ICGNC |
| **Organizing** | Safe and Responsible AI Workshop, GWU (Sep 2024), with the FAA, DOT, Johns Hopkins APL, MIT Lincoln Lab, and NIST |

## TECHNICAL SKILLS

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|---|---|
| **Programming** | Python, C/C++, MATLAB/Simulink; Linux, Git |
| **Machine learning** | PyTorch, JAX, NumPy, scikit-learn; Hugging Face; LoRA/SFT/GRPO fine-tuning of LLMs; flow matching and diffusion models |
| **Simulation, robotics** | BlueSky ATM simulator, ROS 2, Gazebo |
| **Languages** | French (native), English (fluent, C2), Chinese (intermediate, B1) |

## LEADERSHIP AND OUTREACH

**Secretary General, Tsinghua University African Student Association** May 2022 – May 2024
**R&D Engineer, Tsinghua AI International Student Association** Sep 2022 – May 2023

## REFERENCES

**Prof. Peng Wei** (Ph.D. advisor)
Mechanical and Aerospace Eng., George Washington Univ.
pwei@gwu.edu

**Prof. Alexandre Bayen** (Berkeley host)
EECS, University of California, Berkeley
bayen@berkeley.edu